

4.1

3-rozměrnou regresi

$v \in \mathbb{R}^3$.

$$z = ax + by + c$$

$$(6, 0, 1), (1, 0, 1), (0, 1, 1), \\ (1, 1, 2), (1, -1, 0), (-1, 1, 1).$$

$\Rightarrow$ využijte rovnici

$$M^T M \begin{pmatrix} \hat{a} \\ \hat{b} \\ \hat{c} \end{pmatrix} = M^T v.$$

a potom

$$M = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

$$M^T M = \begin{pmatrix} 4 & -2 & 2 \\ -2 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}$$

a

$$M^T \cdot \vec{v} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}.$$

Ted':

$$M^T M \cdot \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix} = M^T \cdot \vec{v}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 3 \\ 2 & 2 & 6 & 6 \end{array} \right) \text{ elementární úpravy}$$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1/3 \\ 2/3 \\ 2/3 \end{pmatrix}.$$

prožít těchto koeficientů:

$$\begin{aligned} z &= ax + by + z \\ &= \frac{1}{3}x + \frac{2}{3}y + \frac{2}{3} \end{aligned}$$
